

Computing with Python

CS0079

Module 4 Submodule 1



Subtopic 1

User-defined Functions

After this submodule, the learner should be able to:

- Create, test, and run user-defined functions using **def**.
- Understand and apply the function components.
- Construct functions that use default value.
- Write functions that either uses default or return a value.
- Understand programs that apply local variables.



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Functions

- a sequence of instructions that performs a task.

Recall: `input()`, `type()`, `print()` are built-in functions.

Python allows us to also define our own functions.



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Functions: Define Your Own Function

`def()`

- used to make new function will be defined.

Components:

- Function Header
- Function Body
- Descriptive String

```
In [ ]: def function_name(parameter_1, parameter_2, ...):  
        """"  
        Descriptive String (optional but recommended)  
        """"  
        # comments about the instructions  
        instructions  
        return output_parameters (optional)
```



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Functions: Define Your Own Function

Components:

- Function Header
 - starts with a keyword **def**
 - followed by a function name
 - followed by a pair of parentheses with the input parameter inside and ends with a colon (**:**)

Parameters are variables that the user need to supply to a function as input and can be manipulated in the body of the function.



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Functions: Define Your Own Function

Components:

- Function Body
 - where the step-by-step instructions to be executed upon calling the function was placed.
 - it is important to note that the instructions in the body of function must be indented, and in the case of multiple instructions, all the instructions must be of the same indent.
 - at the end of function body, it can return some output. As can be seen on the example, it indicates that part is optional, some functions can return some output or not.



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Functions: Define Your Own Function

Components:

- Descriptive String
 - After defining the function header, an optional descriptive string can be placed in the middle of triple single or triple double quotes.
 - This can be used to describe in more detail the task that will be performed by the function.
 - The descriptive string is also accessed by the command **help()**.



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Functions: Define Your Own Function

Example:

An example of a function that asks for a name and prints **hello** + the input name. The function must be called first, for the instructions to be executed.

```
In [1]: def hello_name(name):  
        '''  
        asks for a string `name` then  
        prints hello + name  
        '''  
        print("Hello " + name + "!" )
```

```
In [2]: hello_name("Angelo")  
hello_name("Jeneffer")  
hello_name("John Kenneth")  
hello_name("Elisa")  
  
Hello Angelo!  
Hello Jeneffer!  
Hello John Kenneth!  
Hello Elisa!
```

Different strings was used as input to **hello_name()**.

If the user inputs a number instead of a string, Python will still execute the function until something goes wrong.



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Functions: Functions with default value

```
In [7]: def hello(name="stranger",hour=-1):  
    ...  
    asks for name (of type string)  
    and hour of the day (of type int)  
    then use the inputs for greeting  
    ...  
  
    print("Hello " + name + "!!")  
  
    if hour == -1:  
        print("Good day!")  
    else:  
        if hour <= 1200:  
            print("Good morning!")  
        elif hour <= 1900:  
            print("Good afternoon!")  
        else:  
            print("Good evening!")
```

```
In [8]: hello()
```

```
Hello stranger!  
Good day!
```

```
In [9]: hello("Jasmine")
```

```
Hello Jasmine!  
Good day!
```

```
In [10]: hello(name="Ivy", hour=1400)
```

```
Hello Ivy!  
Good afternoon!
```

```
In [11]: hello(hour=1000, name="Jaclyn")
```

```
Hello Jaclyn!  
Good morning!
```

```
In [12]: hello(hour=2000)
```

```
Hello stranger!  
Good evening!
```



Functions: Functions with default value

```
In [4]: def another_hello(name="stranger"):  
        '''  
        asks for a string `name` then  
        prints hello + name  
  
        if a name is not supplied by user, default name to print is "stranger"  
        '''  
        print("Hello " + name + "!" )
```

```
In [5]: another_hello("Justine")
```

Hello Justine!

```
In [6]: another_hello()
```

Hello stranger!



Functions: Functions with return value

The function simply computes for the area of a rectangle given its length and width.

At the end of the function, the keyword **return** was used to indicate the variable that will be output by the function.

Notice that the output of the function was assigned to a variable.

```
In [13]: def area_of_rectangle(length, width):  
        '''  
        computes for the area of a rectangle  
        given its length and width  
        '''  
        area = length * width  
        return area
```

```
In [14]: area_1 = area_of_rectangle(2,4)  
        area_1
```

```
Out[14]: 8
```

```
In [15]: area_2 = area_of_rectangle(2.3, 5.0)  
        area_2
```

```
Out[15]: 11.5
```



Functions: Functions with return value

Also notice that different data types of input was performed on the first and second call of `area_of_rectangle()`.

As such, checking the type of `area_1`, the value is different from the type of `area_2`.

This only goes to show that Python gives the user freedom to assign variables to different data types.

```
In [16]: type(area_1)
```

```
Out[16]: int
```

```
In [17]: type(area_2)
```

```
Out[17]: float
```

Functions: Local Variables

Question: Can we access the variables that was defined inside functions?
Take **area_of_rectangle()** for example, can we use the variable area even when we already exit the function?

Answer: No.

```
In [18]: area

-----
NameError
<ipython-input
----> 1 area
```

When defining a function, Python creates a separate listing of variables for it. This list of variables can only be accessed during the execution of the function.

Functions: Local Variables

However, the variables that are local to the function is not part of the variables tracked across the whole Python notebook.

```
In [19]: area = "another term for location"  
         area_3 = area_of_rectangle(5,10)  
         area
```

```
Out[19]: 'another term for location'
```



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Exercise:

Write a program that calculates and prints the number of minutes in a year.

Write a program that takes the radius of a sphere (a floating-point number) as input and outputs the sphere's circumference.



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Q&A SESSION





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